B020315(020)

B. Tech. (Third Semester) Examination, Nov.-Dec. 2022

(AICTE Scheme)

(Civil Engg Branch)

BUILDING MATERIALS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

*Note : Attempt all questions. Part (a) of each question is compulsory & carry 4 marks. Attempt any **two** parts from (b), (c) and (d) and carry 8 marks.*

Unit-I

1. (a) What is IS Code for Ennore sand?

(b) What are Bogue compounds and their functions?

What is their role in hydration mechanism in cement?

(c) What is fineness modulus of aggregate ands what is its unit? Explain abrasion test and impact test of aggregates.

(d) What are natural puzzolonas? What are the types of pozzolans and their characteristics?

Unit-II

2. (a) What is True Slump?

(b) How can you test the quality of fresh and hardened concrete?

(c) Write short notes on :

(i) EC of concrete

(ii) Yield strength of concrete

(iii) Flexural strength of concrete

(iv) Creep of concrete

(d) What are the variables in proportioning of concrete mix? What are the criteria that must be used when designing a concrete mix?

Unit-III

3. (a) What is low cost construction materials?

(b) What are the characteristics of :

(i) Particle Board

(ii) Plywood

(c) Which materials waste can be used in construction industry and how?

(d) What is form work and what are requirements of good form work?

Unit-IV

4. (a) Which type of varnish is used on hardwood substances to hide grain defects?

(i) Wax Polish

(ii) Asphalt Varnish

(iii) Flat Varnish

(iv) French Polish

(b) What material is Bitumen? What products are made from bitumen?

(c) How Distempering is different from painting? Which one is better distemper or emulsion and how?

(d) What is the full form of PVC with respect to paint? What are the different ingredients of paint?

Unit-V

5. (a) What is Structural Steel?

(b) What is Aluminium and its alloys? Write down its properties, uses and advantages.

(c) What is glass their types and their uses in civil engineering?

(d) Write short notes on :

(i) Ceramics

(ii) Types of tiles

(iii) Refractories

(iv) Paver block

(v) Uses of PVC